Arista Access Node KR-ACC-24 Optical Transport Failure Resolution SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3043

Description

The Arista access node KR-ACC-24 is equipped with optical transport interfaces supporting SONET/SDH and OTN protocols, facilitating high-speed data transmission over fiber optic cables. In the event of an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) as indicated by Alarm Code ALARM_3043 of severity Minor, this SOP outlines the necessary diagnostic evaluation and resolution procedures to ensure prompt restoration of service. The failure domain is attributed to the susceptibility of the optical interfaces to physical damage, environmental stressors, or manufacturing defects, which can compromise signal integrity and lead to equipment malfunctions.

Pre-requisite Checks

- Prerequisite: Verify that remote hands have been dispatched to KR Puram Access site to provide physical access to the access node., Rationale: Physical access is essential for inspecting and replacing faulty components.

- Prerequisite: Check the status of redundant links to ensure that they are operational and can take over traffic in case of a primary link failure., Rationale: Redundant links provide a backup path for traffic in case of primary link failure, ensuring minimal service disruption.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to high temperature or humidity	High
Fiber cuts or damage during site maintenance	Medium
Rx/Tx power loss due to software configuration issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface gigabitEthernet 0/0/1, Expected Output: GigabitEthernet0/0/1 is up, line protocol is up, Description: This command verifies the operational status of the optical interface and checks for any errors or warnings.

Command: show optics gigabitEthernet 0/0/1, Expected Output: Optics status: Normal, Description: This command checks the optical interface status, including temperature, power, and laser status.

Command: show log | include ALARM_3043, Expected Output: ALARM_3043: Optical Transport Failure, Description: This command retrieves the system logs to verify the occurrence of the Alarm Code ALARM_3043 and the corresponding failure reason.
```

Resolution Steps

Step 1: Step: Inspect and replace the faulty SFP/XFP, Command: install new SFP/XFP in slot 0/0/1, Description: Perform a visual inspection of the SFP/XFP to identify any signs of physical damage or degradation. Replace the faulty component with a new one if necessary.

Step 2: Step: Verify fiber connections and inspect for any signs of damage, Command: show interface gigabitEthernet 0/0/1 | include fiber, Description: Inspect the fiber connections to ensure they are securely connected and not damaged. Verify that the fiber connections are clean and free of any debris.

Step 3: Step: Configure the Rx/Tx power levels, Command: configure terminal; interface gigabitEthernet 0/0/1; power level rx [value]; power level tx [value]; end, Description: Adjust the Rx/Tx power levels to ensure optimal signal strength and integrity.

Post-Check Verification

```
Command: show interface gigabitEthernet 0/0/1, Expected Output: GigabitEthernet0/0/1 is up, line protocol is up, Description: Verify that the optical interface is operational and there are no errors or warnings.

Command: show optics gigabitEthernet 0/0/1, Expected Output: Optics status: Normal, Description: Verify that the optical interface status is normal, including temperature, power, and laser status.

Command: show log | include ALARM_3043, Expected Output: , Description: Verify that the Alarm Code ALARM_3043 is no longer present in the system logs.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: remove SFP/XFP from slot 0/0/1, Description: Restore the original SFP/XFP if the replacement did not resolve the issue.

- Step: Revert Rx/Tx power level configuration, Command: configure terminal; interface gigabitEthernet 0/0/1; no power level rx [value]; no power level tx [value]; end, Description: Revert the Rx/Tx power level configuration to its original state if the adjustment did not resolve the issue.